Technische Hochschule Rosenheim

Modul - Fortgeschrittene Programmierkonzepte

Bachelor Informatik

01 - Einführung

Prof. Dr. Marcel Tilly

Fakultät für Informatik, Cloud Computing

Tiny logo competition!

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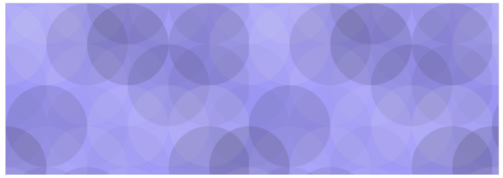
Let's try to find a new logo and replace the boring one...



Price

- Send your proposals to me until Friday!
- Voting starts Saturday ...
- ... until Monday eol (=end of lecture)!

Organizational Things

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Material:

- Lectures
  - Learning Campus: https://learning-campus.th-rosenheim.de/course/view.php?id=1482
  - Online: https://matworx.org/index.html?inf-fpk/README
- GitLab:
  - Examples: https://inf-git.fh-rosenheim.de/inf-fpk/hsro-inf-fpk-github-io/-/tree/master/examples?ref_type=heads
  - Exercises: https://inf-git.fh-rosenheim.de/inf-fpk

> **IMPORTANT: Slides, Script, Exercises ...in English!** ☺

- Subscribe under **Fortgeschrittene Programmierkonzepte (INF-B3), WiSe24/25** (Selbsteinschreibung ohne Schlüssel!)

Lecture - Timetable

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- **Date**: Every Monday, 09:45- 11:15 in R0.03
- **Exercises** in S1.29: Monday, 3./4./5.
  - Group 1: 11:45-13:15 (in presence)
  - Group 2: 13:45-15:15 (in presence)
  - **LB Elvis Helbrandt**
- **Group selection in LC** (now open!)
- Communication via Discord (**#fpk**)

Leistungsnachweis

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Klausur!

- schriftliche Prüfung (SP, 90 Minuten)
- am Ende des Semesters
- erlaubt ist ein Buch mit ISBN Nummer
- Anmeldung über OSC
- Was kommt dran?
  - Alles was in der Vorlesung dran war!
  - Coding on paper!

Projects

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I would like to work with you on small coding projects!

- Form a team (2-4 members)
- Create **your own idea** and drive your **own** project
- Improve your coding skills
- Collaborate with **me**
- Use GitLab or Github
- Show at the end your result (if you want!)

Ideas: Footballmanager, lightweight editor, notes, TicTacToe, Trainingsplaner ...

Lernziele

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Aus dem Modulhandbuch

Die Studierenden …

- … **vertiefen** ihre Kenntnisse in der objektorientierten Programmierung am Beispiel einer geeigneten Programmiersprache (hier: Java!)
- … können die Möglichkeiten und Gefahren der objektorientierten Programmierung **beurteilen**.
- … **sind befähigt**, alle wichtigen Programmierkonzepte für das Programmieren im Großen im Sinne der Komponentenorientierung anzuwenden.
- … **erarbeiten sich die Grundlagen** der funktionalen Programmierung und deren Anwendungsgebiete.

Review

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Programmieren 1

- Imperative programming in C
- Constants, Variables, Expressions, Functions, I/O
- Data structures (fields, arrays, lists)
- Pointer ☺

Review

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Programmieren 1

- Imperative programming in C
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Programmieren 2 (OOP)

- Objekt-oriented Programming (OOP) in Java
- Classes and Objects
- Interfaces and Inheritance
- Exception handling

Agenda of FPK

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Learning Campus

1. Classes, Interfaces and Inheritance
2. Generics
3. Reflection and Annotations
4. GUI
5. Design Pattern and UML
6. Parallel Processing
7. Functional Programming and Streams

Agenda für heute

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1. *Inform:*
   Use your trusted advisors: chatGPT -- Google -- SO -- Java Docs -- Translate
2. *Memorize:*
   The git version control system (https://git-scm.com/)
3. *Automate:*
   The Gradle build tool (https://gradle.org/)
4. *Organize:*
   The IntelliJ IDEA (https://www.jetbrains.com/idea/)
5. *(Optional) Collaborate*
   Practice cross-repository pull requests and learn about *continuous integration*

   Code with me in IntelliJ

SO = stackoverflow

Exercise 1

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Which version control systems do you know?

Version Control

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Git

- Git is a distributed version-control system for tracking changes in source code during software development.
- It is designed for coordinating work among programmers, but it can be used to track changes in any set of files.
- Git is the *de-facto* state of the art version control system.
  - Some of you might remember CVS (concurrent versions system) or subversion.
  - Generally speaking, you should always use a version control system (VCS) when working on code, so you can keep track of changes.
- **Print and laminate**:https://education.github.com/git-cheat-sheet-education.pdf
- For the more visual: http://ndpsoftware.com/git-cheatsheet.html
- If you run into a mess (and you will): https://gandrille.github.io/tech-notes/GIT/git-pretty.pdf

Exercise 2

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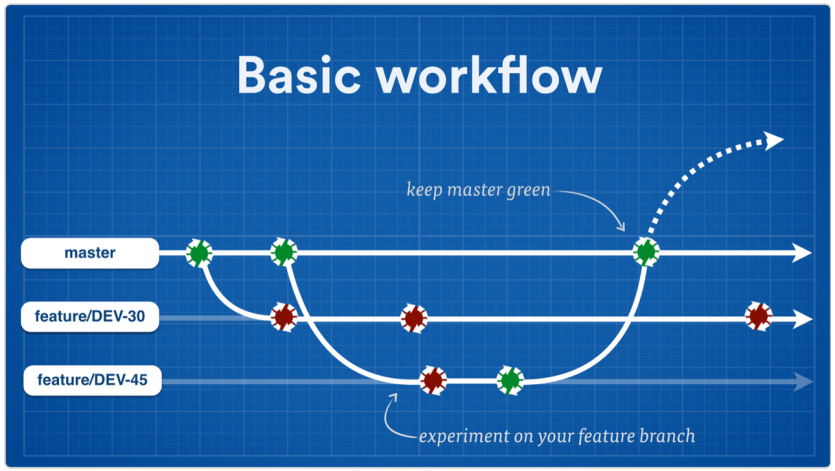
Why is versioning so important?

Git and feature branches

https://www.atlassian.com/continuous-delivery/continuous-delivery-workflows-with-feature-branching-and-gitflow



- Git Guide: https://rogerdudler.github.io/git-guide/
- Git != GitHub

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The Gradle Build Tool (GBT)

Gradle is an open-source build-automation system that builds upon the concepts of Apache Ant and Apache Maven and introduces a Groovy-based domain-specific language instead of the XML form used by Apache Maven for declaring the project configuration.

https://gradle.org

- `gradle init --type java-application` to bootstrap a project
- `./gradlew build` to use the Gradle wrapper to be independent of locally installed Gradle
- `apply plugin: 'eclipse'` and `./gradlew eclipse` to generate Eclipse project files
- `apply plugin: 'idea'` and `./gradlew idea` to generate IntelliJ files (note: these are file-based project descriptions, not the new directory based `.idea/*`)

Exercise 3

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Why is automation so important?

Intellij

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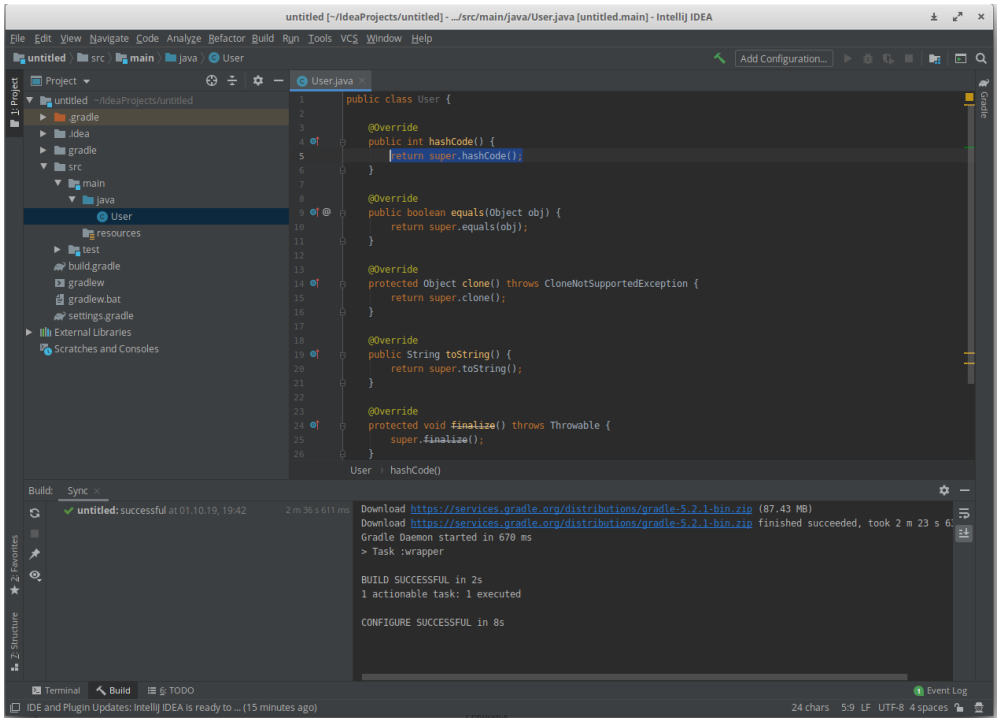
https://www.jetbrains.com/idea/



Collaborate

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- Collaboration means splitting the work
- Teamwork means working together
- Use feature branches and automated tests (JUnit)
- Use automated build and test runner

=> **CI/CD**

- CI and CD are two acronyms that are often mentioned when people talk about modern development practices.
- CI is straightforward and stands for **continuous integration**: A practice that focuses on making preparing a release easier.
- CD can either mean **continuous delivery** or **continuous deployment**.

Travis CI

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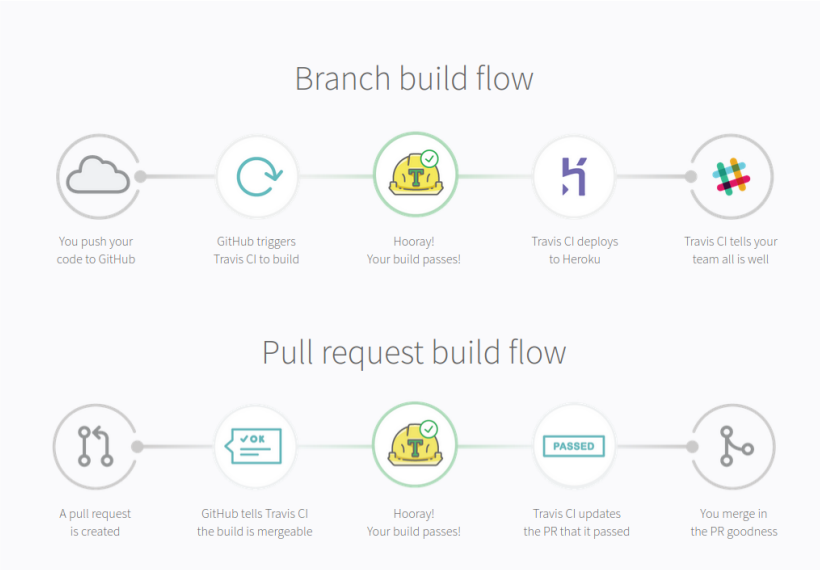
Travis CI is a hosted continuous integration service used to build and test software projects hosted at GitHub. Travis CI provides various paid plan for private projects, and a free plan for open source. (https://travis-ci.org/)



Jenkins

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- An open source automation server
- Jenkins provides hundreds of plugins to support building, deploying and automating any project.
- https://www.jenkins.io/

GitLab

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- GitLab is a web-based DevOps lifecycle tool that provides a Git-repository manager providing wiki, issue-tracking and continuous integration and deployment pipeline features, using an open-source license, developed by GitLab Inc.
- https://about.gitlab.com/ and I am using it for the lecture: https://inf-git.fh-rosenheim.de/inf-fpk

.gitlab-ci.yml:

```
image: gradle:8.3.0-jdk20

stages:
  - build

before_script:
  - echo pwd # debug

build:
  stage: build
  script:
    - ./gradlew test

after_script:
  - echo "End CI"
```

Comparison

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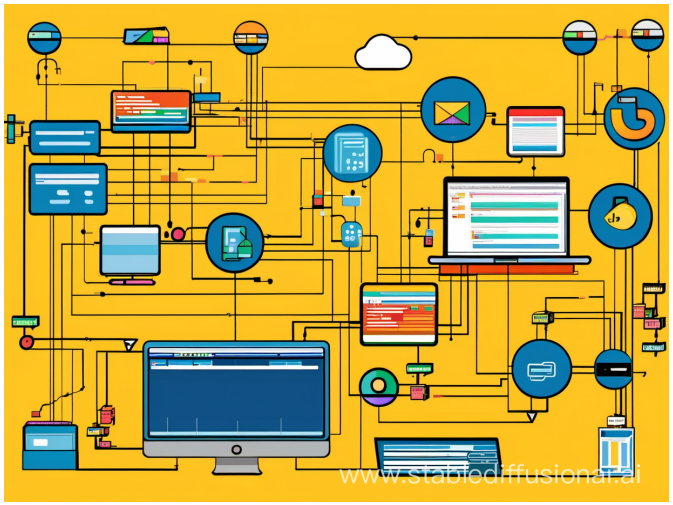
- https://stackshare.io/stackups/gitlab-ci-vs-jenkins-vs-travis-ci#pros
- We are going to use **GitLab**!
- We (TH Rosenheim) are hosting our own server: https://inf-git.fh-rosenheim.de
- Ensure that you have access!



Finally

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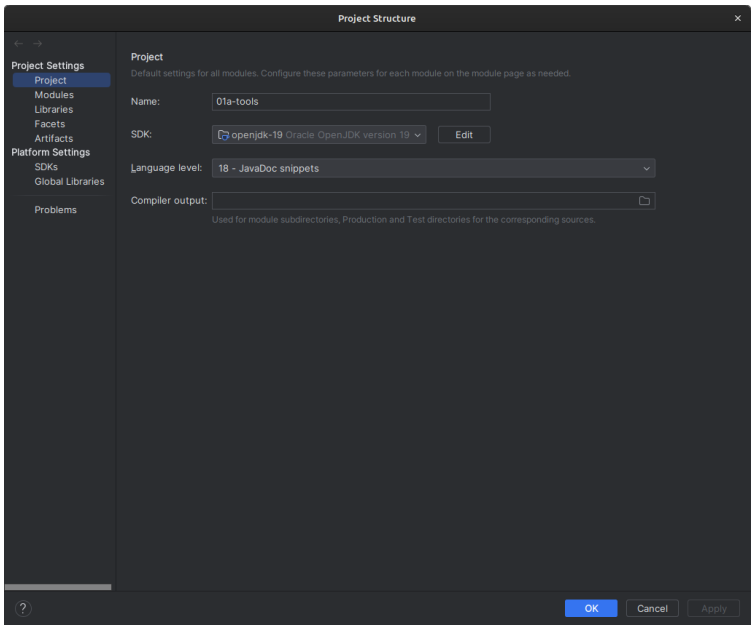
Which versions do we use?

- Java/JDK: JDK 19.0.2 or openJDK 19
- Java Version: Java 18 (JavaDoc Snippets)
- Gradle: 8.3
- Git: 2.42.0



Summary

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- We will look into advanced programming concepts in Java (starting next week!)
- We will to move towards professional software engineering tools (starting today!)
  - Git
  - IntelliJ Idea
  - Gradle
  - GitLab
- Let's try to have fun!

Final thought!